PL5 — Broken-Wing Put Butterfly (SPX)

Presentation report · Prop Desk Quant · 2026-06

QuantConnect backtest · Jun 2021 - Jun 2026 (5 years) · enters every Friday 10:00 ET

WHAT IT IS — a defined-risk put structure with a crash-convex tail.

Per unit: +1 put @ -30 delta / -2 puts @ -18 delta / +2 puts @ -3 delta.

Small net debit. Profit 'tent' if the market drifts down moderately; defined max loss in a valley near the lower strike; the position turns convex again far below (the crash tail).

WHAT WE TESTED — 4 holding horizons: 21 / 28 / 45 / 60 DTE, each over the full 5 years.

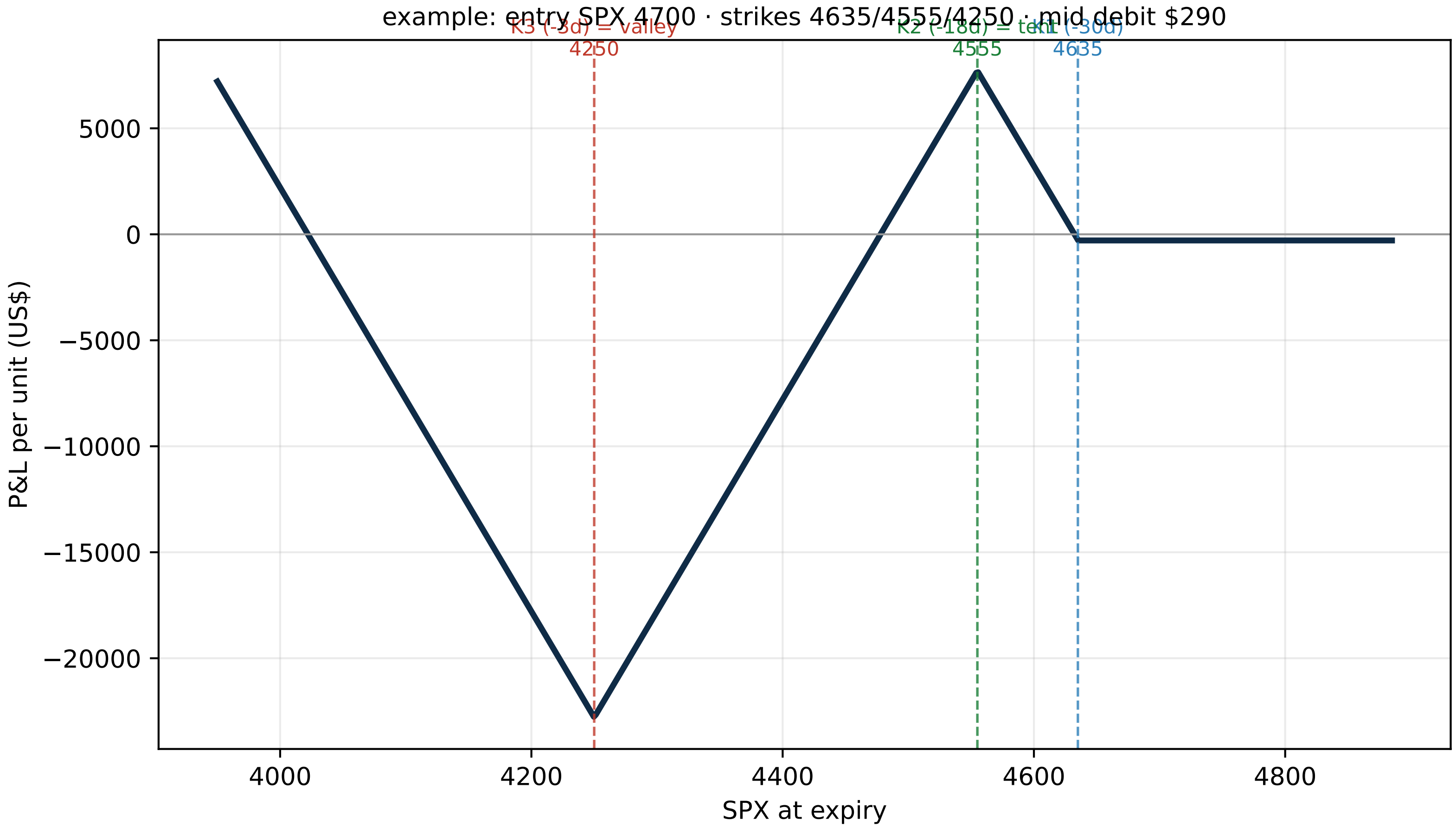
KEY TAKEAWAYS

1. The result is DECIDED BY EXECUTION, not by the structure being right or wrong.
   Same backtest, 28 DTE: $+131,102 at mid -> $+4,087 if every leg is crossed at full bid/ask. The 5-leg entry spread is what moves the number.
2. Traded properly (as ONE combo at a limit price), realistic fill is close to mid:
   21-28 DTE positive ($+66,642 / $+99,348 at a 25% fill), 60 DTE not.
3. Holding to expiry is the WORST exit — a 'valley' reforms in the last days (CZ was right).
4. Real risk = a directional crash that lands in the valley (e.g. the 2025 tariff crash).

All P&L at MID pricing. Methodology and the leg-vs-combo point are explained on pages 2-3.

2. How the structure works

expiry payoff of one unit — where it makes and loses money



Reading the payoff (left to right):

• Above K1 -> all puts expire worthless: you lose only the small debit paid.

• Around K2 -> the 'tent': maximum profit if the market drifts down moderately.

• Around K3 -> the 'valley': the defined MAXIMUM LOSS (a crash that stops right here hurts).

• Far below K3 -> the +2 deep puts dominate again: convex payoff in a deep crash (the tail).

Entry: every Friday 10:00 ET, strikes chosen by delta (-30 / -18 / -3). Defined risk, no naked legs.

3. How we backtested it (and why it's trustworthy)

methodology — full transparency

We deliberately removed every source of fake P&L. What CZ should know:

ENGINE — synthetic tracking.

We subscribe to the real option chain and compute P&L analytically from the actual bid/ask of each leg. We do NOT route orders through QuantConnect's fill/margin engine, which fabricated phantom P&L on multi-leg structures. So the numbers reflect real quoted prices, nothing invented.

SETTLEMENT — exact, no spread.

SPX options are European, cash-settled. At expiry the position settles at the official intrinsic value — there is NO exit spread. Verified on page 8 (theoretical payoff == recorded P&L).

PRICING — mid is the standard; we also stress the fill.

Entry, marking and exits all use MID consistently. Because settlement has no spread, the ONLY thing separating 'mid' from 'bid/ask' on a hold is the ENTRY price of the 5 legs.

LEG-BY-LEG vs COMBO — the most important nuance for CZ.

Our pessimistic case sums each leg at its OWN bid/ask (as if you sent 5 separate market orders). A butterfly is never traded that way — it is sent as ONE combo at a single net limit. The combo's net spread is far tighter than the sum of the legs (the legs' risks offset; the MM hedges the package). Since mids are additive, our 'mid' number IS the combo's net mid — the price you would actually aim for. So reality sits close to mid, NOT at the leg-by-leg worst case.

VERIFIED at minute resolution.

We suspected the wide tail spread was a stale hourly-quote artifact. We re-ran at minute data: the entry spread is IDENTICAL (1.000x). It is the real far-OTM tail market, not a data glitch.

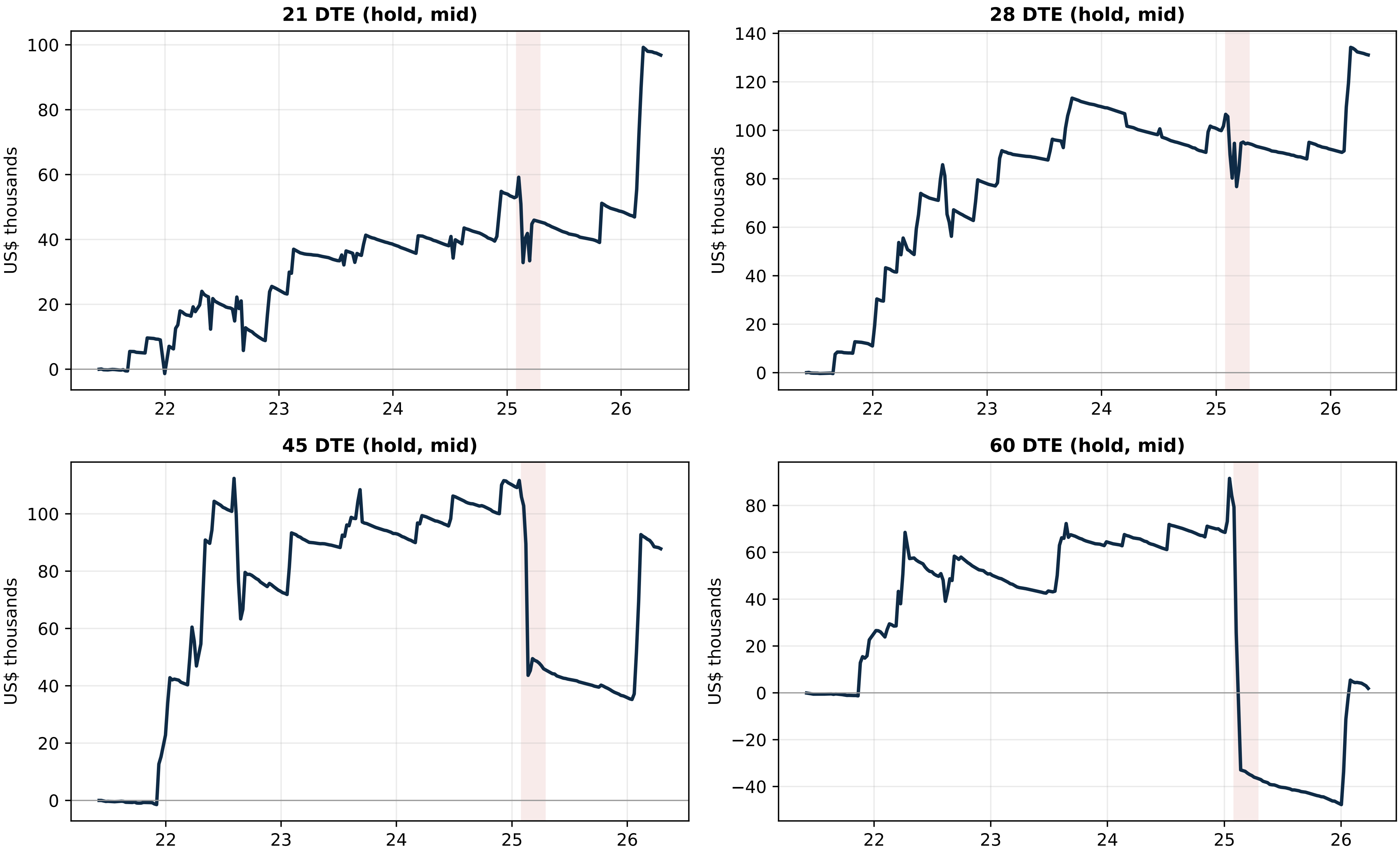
4. The key finding — sensitivity to entry fill (hold to expiry)

net P&L over 5 years · realistic combo execution = near 'mid'; far-right = 5 market orders (unrealistic)

Horizon	Combo mid	25% of spread	50% of spread	Leg-by-leg (worst)
21 DTE	$+96,831	**$+66,642**	$+36,453	$-23,924
28 DTE	$+131,102	**$+99,348**	$+67,594	$+4,087
45 DTE	$+87,811	**$+49,419**	$+11,026	$-65,759
60 DTE	$+1,897	**$-47,219**	$-96,336	$-194,568

HOW TO READ THIS: the four columns are the SAME trades priced at four entry-fill assumptions. 'Combo mid' = you fill the whole butterfly at its net mid (a patient limit order). 'Leg-by-leg (worst)' = you cross the full bid/ask on every one of the 5 legs separately (nobody trades a fly this way). The truth for a combo order sits on the LEFT side, near 25%. The spread is real (verified at minute data) and lives almost entirely in the illiquid -3 delta tail. CONCLUSION: PL5 is execution-bound; traded as a combo at a limit, 21-28 DTE is positive, 45 DTE marginal, 60 DTE negative (its wing is so wide the valley is too deep).

5. Cumulative P&L (mid) by horizon — full 5 years



Shaded band = the 2025 tariff crash. All curves are MID; net of the entry spread the slopes flatten (see page 4).

6. Exit rule x horizon — the full matrix (net P&L, mid)

every close rule x every horizon · 5-year net at mid

Close rule	21 DTE	28 DTE	45 DTE	60 DTE
Hold to expiry	$+96,831	$+131,102	$+87,811	$+1,897
Exit at 30 DTE left	-	-	$+21,105	$+40,810
Exit at 21 DTE left	-	$+8,485	$+18,008	$+52,508
Exit at 14 DTE left	$+10,600	$+20,035	$+31,242	$+85,928
Exit at 10 DTE left	$+17,725	$+22,250	$+64,602	$+81,588
Exit at 7 DTE left	$+13,490	$+17,372	$+73,268	$+63,975
Exit at 5 DTE left	$+22,955	$+20,355	$+60,300	$+77,615
Exit at 3 DTE left	$+12,508	$+42,535	$+118,272	$+117,478

KEY INSIGHT — the horizon ranking FLIPS between holding and exiting early: HELD TO EXPIRY, short horizons win (28 DTE best; 60 DTE collapses to +$2k, its valley is deepest at expiry). With EARLY EXIT (how PL5 is actually traded) LONG horizons win (45/60 DTE; e.g. exit 3 days before expiry: 45 DTE +$118k vs 21 DTE +$13k). Longer DTE gives the downside tent more time to form, and leaving before the final days avoids the valley reforming (60 DTE: +$2k held vs +$117k if exited 3 days early). All at mid; realistic combo fill scales every cell down similarly, so the relative ranking holds.

7. A real worst case — the 2025 tariff crash (60 DTE)

trade 2025-02-07 -> 2025-04-04 · entry SPX 6099 · strikes 5975/5790/5050

Exit point	SPX that day	P&L (mid)
14 DTE left (2025-03-21)	5668	$+475
10 DTE left (2025-03-25)	5777	$+5,120
7 DTE left (2025-03-28)	5581	$+4,065
3 DTE left (2025-04-01)	5633	$-1,765
SETTLE (2025-04-04)	5075	$-53,541

SPX collapsed in the final days to 5075, right at K3 (5050) = the valley = max loss. A holder took $-53,541; exiting a week earlier was near flat. This is the genuine risk of PL5: a directional crash that lands in the valley. The loss is defined, but it is large and, with overlapping weekly entries, can cluster. This is real risk, at mid — not an execution artifact.

8. Data verification

methodological rigor

Why these numbers are real, not fabricated:

1) Settlement = exact intrinsic at the SPX official close (cash-settled, European) — no spread, no phantom fills. Check on the 3 worst 60 DTE trades (theoretical payoff vs recorded hold):

    2025-02-07: SPX 5075 -> theoretical payoff $-52,976 (recorded hold $-53,541)
    2025-02-21: SPX 5361 -> theoretical payoff $-28,880 (recorded hold $-29,455)
    2025-02-14: SPX 5361 -> theoretical payoff $-28,880 (recorded hold $-29,245)

2) Strikes verified per trade (chosen at -30 / -18 / -3 delta).

3) MID pricing applied consistently across entry, marking and every exit rule.

4) Entry spread VERIFIED at minute resolution = identical to hourly (1.000x): the tail spread is the real market, not a stale-quote artifact (so the worst case is a legitimate bound).

5) Every trade is auditable in the app (date, strikes, DTE, debit, VIX, spot, exits, P&L).

9. Verdict

honest conclusion

PL5 is NOT a 'hold to expiry' trade — the valley at the end destroys it. And it is NOT a market-order trade — it must be worked as a combo at a limit price.

Realistic execution (combo fill, ~25-50% of the leg-by-leg spread), 5-year net P&L:

- 21 DTE: $+66,642 (25%) / $+36,453 (50%) -> positive
- 28 DTE: $+99,348 (25%) / $+67,594 (50%) -> best, positive
- 45 DTE: $+49,419 (25%) / $+11,026 (50%) -> marginal
- 60 DTE: $-47,219 (25%) / $-96,336 (50%) -> negative

Honest caveats:

- Execution is THE decisive factor (5 legs, illiquid -3d tail). Fill discipline = the edge.
- Real tail risk: a crash into the valley = a large, defined loss; can cluster with overlapping weekly entries (the test enters every Friday with no concurrency cap).

Suggested next step: forward-test 28 DTE with early exit + a concurrency rule, and — above all — measure the REAL combo fill on the tail, which is the single factor that decides this strategy.